QMS Modules – Life Sciences (Supply Chain OS)

API & Raw Material Manufacturing Context

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1. In-Process Quality

What it does

Monitors and records quality checks **during** the manufacturing process — before the product is finished.

Why it exists

To catch quality issues **early** during production, preventing defective batches from moving forward.

End-to-End Workflow

1. Production starts a batch (e.g., API synthesis)
2. At defined checkpoints, samples are collected
3. QA Executive checks parameters (pH, temperature, particle size, moisture)
4. Results are recorded in the system
5. If within limits → production continues
6. If out of limits → alert raised → Deviation triggered

Examples

- **API Manufacturing:** Checking the reaction temperature every 2 hours during API synthesis
- **Raw Material:** Verifying moisture content of an incoming raw material before it enters the blending process

Role Interaction

Role	Action
QA Officer	Sets acceptable limits, reviews results, approves/rejects
Production Manager	Ensures checks happen on time, acts on alerts

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2. Deviation Management

What it does

Records and manages any **unplanned departure** from approved procedures, specifications, or standards.

Why it exists

Regulations (FDA, GMP) require all deviations to be documented and investigated to ensure product safety.

End-to-End Workflow

1. An unexpected event occurs during production
2. Operator/Production Manager raises a Deviation record
3. QA Officer reviews and classifies (Minor / Major / Critical)
4. Root cause investigation is performed
5. Impact assessment done (Does it affect product quality?)
6. Decision: Accept batch / Reject / Rework
7. CAPA initiated if needed
8. Deviation closed after review and approval

Examples

- **API Manufacturing:** Temperature of reactor exceeded set limit by 5°C for 10 minutes
- **Raw Material:** Supplier delivered raw material 2 days late, stored in non-standard conditions

Role Interaction

Role	Action
QA Officer	Classifies deviation, investigates root cause, approves closure
Production Manager	Raises deviation, provides context, implements corrective steps

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3. CAPA (Corrective and Preventive Actions)

What it does

A structured system to **fix existing problems (Corrective)** and **prevent future ones (Preventive)**.

Why it exists

Core requirement of GMP and ISO standards. Ensures problems don't repeat and systemic risks are eliminated.

End-to-End Workflow

1. Problem identified (from Deviation, Audit, Complaint, etc.)
2. CAPA record created
3. Root cause analysis performed (5-Why, Fishbone, etc.)
4. Corrective Action defined (fix the current issue)
5. Preventive Action defined (stop recurrence)
6. Actions assigned to responsible persons with deadlines
7. Actions implemented and verified
8. Effectiveness check done
9. CAPA closed

Examples

- **API Manufacturing (Corrective):** Re-calibrate reactor temperature sensor that caused deviation
- **API Manufacturing (Preventive):** Add automated alerts when temperature approaches limit
- **Raw Material (Preventive):** Add vendor qualification step to prevent substandard material

Role Interaction

Role	Action
QA Officer	Approves CAPA plan, verifies effectiveness, closes CAPA
Production Manager	Implements actions, trains team, provides evidence

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4. In-Product Quality

What it does

Quality testing performed on the **finished product** before it is released for sale/distribution.

Why it exists

Ensures the final product meets all regulatory and customer specifications before it leaves the facility.

End-to-End Workflow

1. Batch manufacturing is complete
2. Samples sent to QC lab
3. Tests performed: Assay, Purity, Dissolution, Microbial, etc.
4. Results compared against approved specifications
5. QA reviews batch records + test results
6. Batch Released (Pass) or Rejected (Fail)
7. Certificate of Analysis (CoA) generated for release

Examples

- **API Manufacturing:** Purity test of final API batch must be ≥99.5%
- **Raw Material:** Testing incoming excipient for identity, purity, and microbial load

Role Interaction

Role	Action
QA Officer	Reviews test results, approves/rejects batch, signs CoA
Production Manager	Ensures batch records are complete, coordinates with QC lab

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5. Product Complaints

What it does

Captures and manages **complaints received from customers or end users** about product quality or safety.

Why it exists

Regulatory requirement. Complaints can signal systemic issues or product failures that need investigation.

End-to-End Workflow

1. Complaint received (from customer, distributor, or market)
2. Complaint logged in system with details
3. Initial assessment: Is it a safety issue?
4. Investigation: Batch records reviewed, retain samples tested
5. Root cause identified
6. Response sent to complainant
7. CAPA triggered if systemic issue found
8. Regulatory reporting if required (e.g., adverse event)

Examples

- **API Manufacturing:** Customer reports API batch has unusual color, not matching spec
- **Raw Material:** Manufacturer reports that a raw material caused poor yield in their process

Role Interaction

Role	Action
QA Officer	Logs complaint, leads investigation, communicates with customer
Production Manager	Provides batch records, checks if issue is production-related

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6. Recall Management

What it does

Manages the process of **withdrawing a product from the market** due to a safety, quality, or compliance issue.

Why it exists

To protect patient/consumer safety. Regulatory agencies (FDA, EMA) mandate recall procedures.

End-to-End Workflow

1. Trigger identified: failed test, complaint, adverse event, regulatory alert
2. Risk assessment: Which batches are affected?
3. Recall classification (Class I / II / III based on risk)
4. Notify regulatory authority
5. Notify distributors, customers, hospitals
6. Retrieve affected product
7. Quarantine and destroy/rework
8. Root cause investigation
9. CAPA initiated
10. Recall closed and reported

Examples

- **API Manufacturing:** API batch found to have microbial contamination post-release
- **Raw Material:** A raw material supplier issues a recall due to wrong labeling

Role Interaction

Role	Action
QA Officer	Initiates recall, notifies regulators, tracks returned product
Production Manager	Identifies affected batches, coordinates logistics of retrieval

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7. Adverse Event Reporting

What it does

Records and reports **unexpected harmful or unintended effects** associated with a product.

Why it exists

A legal and ethical obligation. Regulatory agencies (FDA MedWatch, EMA EudraVigilance) require timely reporting to protect public health.

End-to-End Workflow

1. Adverse event identified (from complaint, clinical report, or post-market surveillance)
2. Event logged: product details, patient info, description of event
3. Medical assessment: Is it related to the product?
4. Causality assessment performed
5. Regulatory report prepared and submitted within required timeline
6. Follow-up investigation done
7. CAPA initiated if product defect is cause
8. Event closed after reporting confirmed

Examples

- **API Manufacturing:** A hospital reports that patients showed unexpected allergic reactions to a drug containing your API
- **Raw Material:** A contaminant in a raw material caused adverse reactions in end users

Role Interaction

Role	Action
QA Officer	Assesses event, prepares regulatory submission, tracks timelines
Production Manager	Provides manufacturing data, checks batch history for potential cause

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Summary Table

Module	Purpose	Triggered By
In-Process Quality	Monitor during production	Scheduled checkpoints
Deviation Management	Record unplanned events	Any departure from procedure
CAPA	Fix & prevent quality issues	Deviation, Audit, Complaint
In-Product Quality	Release testing of finished product	End of batch
Product Complaints	Handle customer quality complaints	Customer/Market feedback
Recall Management	Withdraw unsafe products	Safety/Quality failure
Adverse Event Reporting	Report harmful product effects	Patient/User safety event